

Experiment 1: Multi-Locator Login Test

Module	Automation with Selenium
Application	SauceDemo (https://www.saucedemo.com/)
Objective	Locate and use form controls with ID, NAME, and XPATH selectors, then verify the destination URL.
Expected result	Successful login redirects to /inventory.html.

1. Test objective

Open the SauceDemo login page, identify the username, password, and login controls with three different locator strategies, submit valid demo credentials, and assert that the browser reaches the inventory page.

2. Prerequisites

Python 3, Selenium WebDriver, Google Chrome, and network access. The credentials below are public demo credentials supplied by SauceDemo.

3. Code implementation

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

VIEW_DELAY = 2

driver = webdriver.Chrome()
wait = WebDriverWait(driver, 15)
try:
    driver.get("https://www.saucedemo.com/")
    time.sleep(VIEW_DELAY)

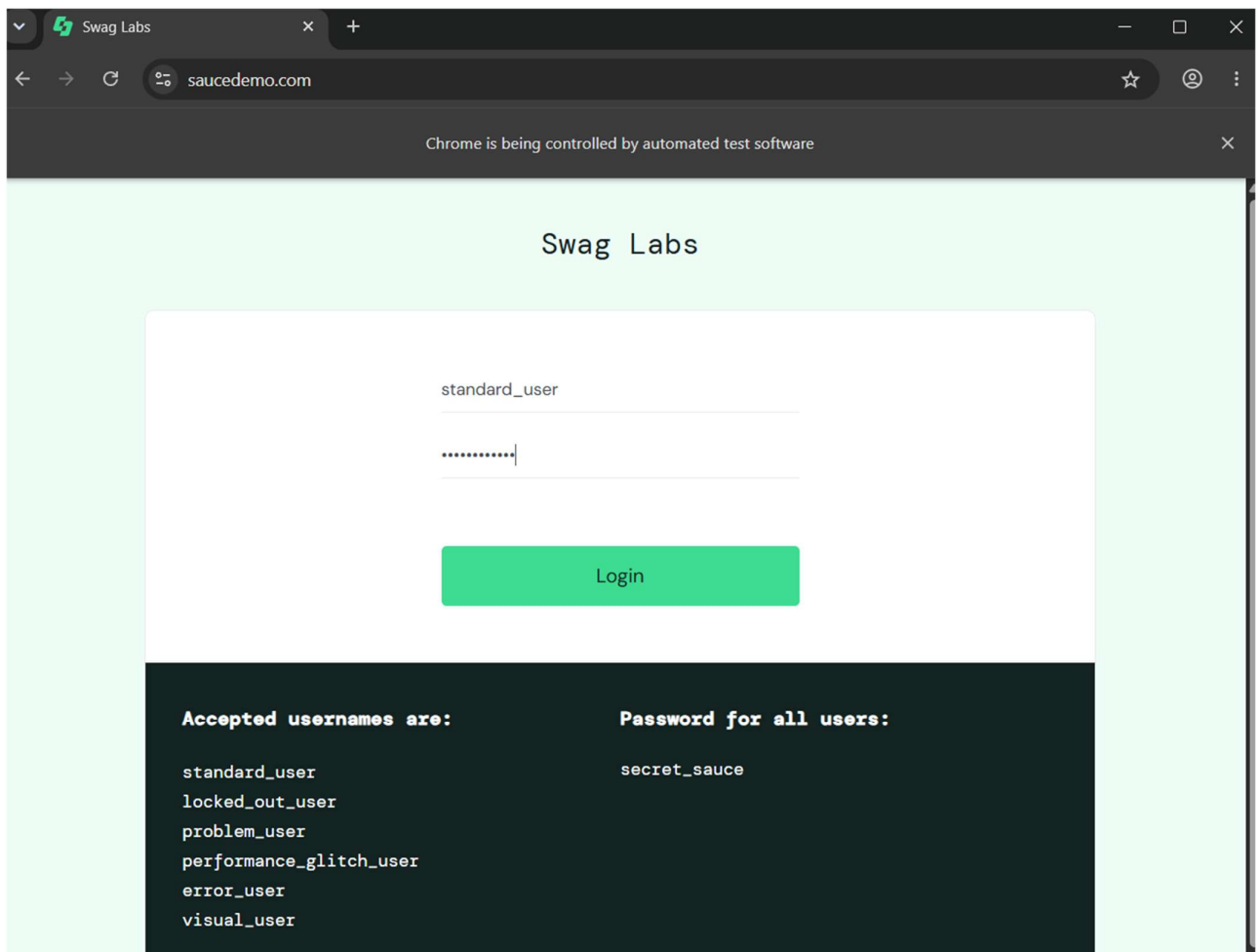
    username = wait.until(EC.presence_of_element_located((By.ID, "user-name")))
    password = driver.find_element(By.NAME, "password")
    login = driver.find_element(By.XPATH, "//*[@id='login-button']")
    username.send_keys("standard_user")
    password.send_keys("secret_sauce")
    time.sleep(VIEW_DELAY)
    login.click()
    wait.until(EC.url_contains("/inventory.html"))
    assert "/inventory.html" in driver.current_url
    print("Test Passed: Successfully logged in and verified URL.")
```

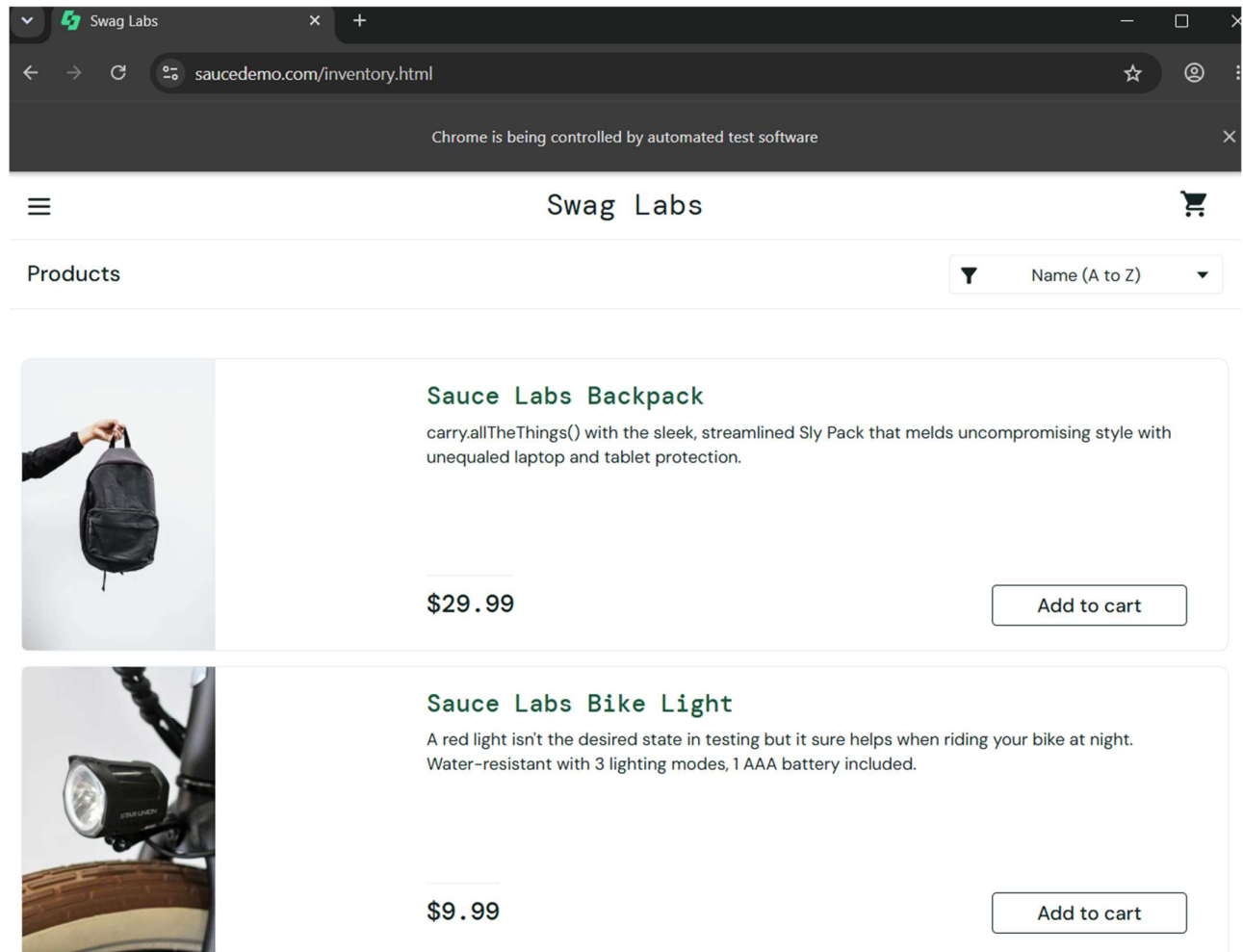
```
time.sleep(VIEW_DELAY)
finally:
    time.sleep(VIEW_DELAY)
    driver.quit()
```

4. Verification record

Check	Observed result
Username located with By.ID	Passed
Password located with By.NAME	Passed
Login button located with By.XPATH	Passed
Login redirect assertion	Passed: https://www.saucedemo.com/inventory.html
Console output	Test Passed: Successfully logged in and verified URL.

5. Screenshot evidence





Screenshot 1 - Login form with standard_user and masked password entered.

Screenshot 2 - Products inventory page after successful login. Both screenshots were captured from the live SauceDemo run and are provided with the deliverable response.

6. Conclusion

The test meets the experiment requirements. ID and NAME selectors identified the fields, XPATH identified the login button, and the URL assertion confirmed successful authentication.